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Influence of landscape change on nearshore fisheries in southern Chile

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Abstract

At least half of the world's population resides in the coastal zone and the livelihoods of billions of people are affected either directly or indirectly by the production and sustainability of nearshore fisheries. Landscape change, specifically development of tree plantations, is accelerating worldwide as developing countries integrate into global markets to sell goods, offer climate-mitigation services (carbon), and/or provide renewable energy. These changes can release excess nutrients into adjacent coastal waters causing eutrophication that alters the structure and function of coastal ecosystems. This study examined the relationship between coastal drainage basin land use/land cover change (LCLUC), specifically development of tree plantations, patterns of chlorophyll-*a* in nearshore coastal waters, and the biological condition of commercially important shellfish, *Concholepas concholepas* (*loco*) in southern Chile. *Locos* (N = 1374) were sampled across 13 watersheds (35 853 km^2) and 42 fisheries management areas (spanning 250 km of coastline). *Locos* harvested from management areas influenced by tree plantations had approximately 30% more endobiont (shell-boring) phoronids, almost twice as many endobiont polychaetes and twice as many epibiont (shell-attaching) barnacles than *locos* from areas in close proximity to watersheds dominated by native forests (15–20% of the watershed). Phoronid infested *locos* from coastal waters adjacent to watersheds with tree plantations were of relatively poor biological condition (smaller and narrower in width) and of reduced market value. Our study suggests that tree plantations result in indirect ecological impacts to coastal fisheries (more nutrients and higher phytoplankton biomass, resulting in smaller, low quality locos), and costs are born by coastal fishers (lower prices for locos). Increases in tree plantations could thus potentially significantly impact coastal fisheries worldwide and such problems should be managed as an interconnected network of land use change, oceanic ecosystems, and economic systems that are considered an integrated socio-ecological system.

Keywords: *Concholepas concholepas*, eutrophication, land-sea interface, shellfish, tree plantations

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Introduction

Over half of the world's population lives in the coastal zone (Cohen *et al.*, 1997) and extensive coastal development continues to be a concern. Intensive coastal development and a growing realization that people's actions on land impact marine systems makes modeling of processes that occur at the land-sea interface and the effects of changing land use critical to coastal planning and the protection of marine biodiversity (Stoms *et al.*, 2005). Such modeling identifies unexpected ecological and socio-economic responses and trade-offs arising from decisions about land use (DeFries *et al.*, 2004).

Understanding the effects of changes in land use in drainage basins becomes especially important in developing, coastal countries. The livelihoods of an estimated 23 million people in these countries depend on healthy and productive nearshore environments (FAO, 2002), and rapid changes in land use at the landscape scale can cause significant harm. In addition to pressures exerted by increases in agriculture, urbanization, and animal production, significant pressures are exerted when land is converted to tree plantations.

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Considerable numbers of tree plantations are being created in the southern Hemisphere (Sedjo, 1999; Simberloff *et al.*, 2010) in response to subsidies (Bull *et al.*, 2006) that foster production of pulp and energy to domestic (Berndes *et al.*, 2003) or global demands (Gwynne, 1996) and, more recently, to mitigate the effects of climate change (Berndes *et al.*, 2003; Putz & Redford, 2010). For example, the areal extent of tree plantations in Chile is extensive, with planted trees often replacing native forest, (Echeverria *et al.*, 2006; Aguayo *et al.*, 2009; Van Holt, 2009; and Simberloff *et al.*, 2010) and with development of commercial plantations during the 20th century having radically altered the historical, terrestrial landscape (Wilson *et al.*, 2005). Further development of tree plantations in the coastal ranges is anticipated (Wilson *et al.*, 2005; Van Holt, 2009), especially in areas with moderately steep slopes (Van Holt, 2009) where increased run-off and subsequent adverse ecological impacts on adjacent coastal waters are of particular concern to both fishers and managers of fisheries.

As land is converted, vegetation patterns (Binkley & Resh, 1999; Farley & Kelly, 2004) and the hydrology of lotic systems (Farley *et al.*, 2005; Iroume *et al.*, 2006) are altered. These changes, in turn, often lead to a more rapid delivery of additional nutrients to nearshore coastal waters, subsequent eutrophication (Nixon, 1995), and negative consequences for floral and faunal communities (Diaz & Rosenberg, 1995; Smith *et al.*, 1999; Anderson *et al.*, 2002; Kemp *et al.*, 2005). For example, suspension feeding bivalves (Risk *et al.*, 1995), barnacles (Gabaev *et al.*, 2005), phoronids (Chang *et al.*, 1992; Belan, 2003), and polychaetes (Beukema, 1991; Crema *et al.*, 1991; Chang *et al.*, 1992; Llanso, 1992; Belan, 2003; Perus & Bonsdorff, 2004) are often more abundant in eutrophic waters. Suspension feeders may, in some cases, ameliorate the short-term effects of increased nutrient delivery manifest as increased production of phytoplankton because their feeding rates scale with food availability (Herman & Scholten, 1990). Nevertheless, longer-term shifts in assemblages may occur (Chauvaud *et al.*, 2000; Grall & Chauvaud, 2002), and such changes might represent reliable indicators of ecosystem transformation due to increased delivery of nutrients.

One signal of unbalanced nutrient cycles may be an increase in bivalves, barnacles, phoronids, and polychaetes on and in various substrates (shells, coral, etc.). Abundances of epibionts (e.g., barnacles on the carapace of a crustacean and other organisms that occupy the surface of another organism) and endobionts (e.g., some polychaetes, phoronids, shell-boring bivalves, and other organisms that bore into shells, coral or other substrates), have been shown to increase with increased phytoplankton production, which is a related common response to increased nutrient delivery (Lescinsky *et al.*, 2002). Excessive numbers of epibionts and endobionts negatively influence the livelihoods of fishers because organisms with heavy burdens grow slower (McDiarmid *et al.*, 2004) and fail to reach maximum weights (Lescinsky *et al.*, 2002), which often leads to lower market prices (Van Holt, 2012). In fact, severely affected organisms may become unmarketable as exemplified by non-native oysters in Chesapeake Bay that became heavily fouled with polychaetes during the summer months (Grabowski *et al.*, 2007). The compromised biological condition of organisms can, in turn, alter the structure and function of food webs (Marcogliese, 2005), perpetuating a cycle of ecological damage and degraded resources. The potential negative effects of changes in landscapes on local fisheries become a particular concern in nations, including Chile, that have implemented innovative and effective strategies to manage fisheries by enforcing spatial restrictions on access to benthic species, especially shellfish (Defeo & Castilla, 2005; Gelcich *et al.*, 2010).

To evaluate the impacts of tree plantation development on water quality in the nearshore waters of Chile and the potential consequences for a key exploited species, *Concholepas concholepas* (Bruguière, 1789; Mollusca, Gastropoda, Muricidae), we compared conditions in 42 fisheries management areas spanning 250 km of coastline and 13 distinct watersheds. Landsat satellite images allowed us to characterize changes in land use/land cover across 35 853 km^2, with a focus on tree plantations. By combining these data with estimates of chlorophyll-*a* concentrations in adjacent coastal waters derived from satellite images and relevant condition metrics for *C. concholepas* (*locos*) collected from the sites, we sought key insights into the links between land use and management of living marine resources. We specifically hypothesized that *locos* located in management areas downstream from watersheds dominated by tree plantations would be burdened with more epibionts and endobionts and exhibit relatively poor condition (lower lipid content in their muscle tissue, lower average weight, and shorter length) when compared with *locos* from other, less disturbed regions.

Methods

Chilean loco fishery

Locos are economically the most important shellfish for artisanal fishers in Chile. Following a collapse of the *loco* fishery in 1990, the Territorial User Rights Fisheries (TURF) (Bernal *et al.*, 1999; Gelcich *et al.*, 2010) management program assigned fishers exclusive fishing rights to benthic resources in parcels of the ocean, which are called Managed Exploitation

Areas for Benthic Resources (MEABR). *Loco* fisheries recovered soon thereafter (Defeo & Castilla, 2005; Gonzalez *et al*., 2006). As the MEABRs were being established, certified divers, fishers, or collectors of nearshore resources formed syndicates and solicited management areas from the Servício Nacional de Pesca (SERNAPESCA). A baseline study of each management area (carried out by nationally certified consultants) provided information on *loco* populations, including, estimates of population size, numbers of *locos* in distinct size classes, and biomass (Bitecma, 2002). The consultants used these data and ancillary ecological data to recommend initial harvest quotas; the number of people in the syndicate had no bearing on the quota. Once the quota was calculated, the syndicate turned it over to SERNAPESCA, who modified it if necessary before it was published. Obtaining an MEABR and harvesting *locos* are group efforts – the president of each syndicate processes the legal paper work and organizes the group.

Loco harvests are tightly regulated, and people transporting *locos* must carry documentation from SERNAPESCA. Fishers cannot harvest *locos* within an area that has not undergone an ecological evaluation, and fishers must pay consultants to assess the *loco* population annually although the government may subsidize this activity. In all surveys, consultants sample $\geq$ 1000 *locos*, document size-classes, and recommend a quota based on their findings. Independent assessments have concluded that *loco* populations are stable within the management areas but subject to overharvesting outside of the designated management areas (Gonzalez *et al*., 2006).

On average, artisanal fishers harvest 3278 tons of *locos* in management areas across administrative regions X and XIV (SERNAP, 2003, 2004, 2005) each year. There are 23 000 artisanal fishers (over a third of Chile's artisanal fishers) in region XIV and the northern portion of region X (Fig. 1), and they fish predominantly in designated management areas along the coast (SERNAP, 2003). During the harvest season (July–August), fishers using hookahs work in teams of two to four to harvest *locos*. The president of the syndicates usually coordinates the sale of *locos* to one or two buyers who subsequently export the shellfish. Artisanal fishers rarely harvest other benthic organisms and this offers some protection for the filter-feeding tunicates, mussels, and barnacles consumed by *locos*, a top carnivore (Stotz *et al*., 2003). We have observed fishers harvesting *picoroco* barnacles (*Austromegabalanus psittacus*) in the south of administrative region X (Fig. 1), where these barnacles are perceived to be abundant.

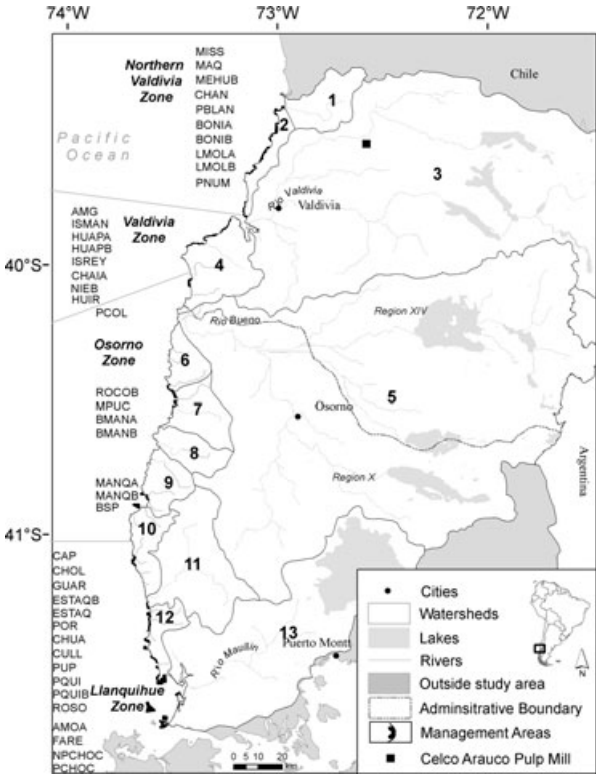

Fig. 1 Management sites where *locos* (*Concholepas concholepas*) were sampled. Numbered watersheds include: Valdivia (watershed # 3), Bueno (5), and Maullín (13) (ranging from 4521 to 15 332 km^2), and ten small watersheds that range from 225 to 1393 km^2 – Lingue (1), Bonifacio (2), Chaihuín (4), Quilhue (6), Contaco (7), Hueyelhue (8), Cholguaco (9) San Luis (10), Llico (11) and Putratrán (12). Abbreviated management areas are grouped according to geographic zones used in the ANOVA anlaysis. The Northern Valdivia zone included: MISS = Mississippi, MAQ = Maiquillahue, MEHUB = Mehuin B, CHAN = Chan Chan, PBLAN = Piedra Blanca, BONIA = Bonifacio A, BONIB = Bonifacio B, LMOLA = Los Molinos A, LMOLB = Los Molinos B, and PNUM = Punta Numpulli. The Valdivia zone included: AMG = Amargos, ISMAN = Isla Mancera, HUAPA = Huape A, HUAPB = Huape B, ISRE = Isla del Rey, CHAIA = Chaihuin A, NIEB = Niebla, and HUIR = Huiro. The Osorno zone included: PCOL = Punta Colun, ROCOB = Roca Covadonga B, MPUC = El Manzano de Pucatrihue, BMANA = Bahia Mansa A, BMANB = Bahia Mansa B, MANQA = Manquemapu A, MANQB = Manquemapu B, and BSP = Bahia San Pedro. The Llanquihue zone included: CAP = Capitanes, CHOL = Cholhuaco, GUAR = Guaguar, ESTAQB = Estaquilla B, ESTAQ = Estaquilla, POR = Punta Ortiga/Puga, CHUA = Chaihuaco, CULL = Cullinco, PUP = Pupelde, PQUI = Punta Quillahua~~e~~, PQUIB = Punta Quillahua B, ROSO = Roca Solitaria, AMOA = Amortajado A, FARE = Farellones, NPCHOC = Norte Punta Chocoi, and PCHOC = Punta Chocoi.

Study design

We employed a cross-sectional comparative approach in our investigations. The validity of comparative research depends, in large part, on selecting sites that are as similar as possible in every respect except for the variable(s) of interest (phytoplankton production and patterns in landscape change). We selected the nearshore rocky intertidal in the Valdivia to Puerto Montt region along Chile's central coast because of broad similarity in oceanographic conditions and zoogeographic patterns. The more northern region along the Chilean coast (near Peru) boasts higher fish abundance and productivity that is driven by major upwelling events that bring cold, nutrient-rich water to the surface and episodic el Niño and la Niña events (Moreno *et al*., 1979; Camus, 2001). The region

south of Puerto Montt (Fig. 1) is dominated by fjords that have more complex geomorphologies and more diverse and abundant assemblages of benthic invertebrates (Fernandez *et al.*, 2000; Lancellotti & Vasquez, 2000).

Within our study region, the northward-flowing Humboldt Current is the main oceanographic influence (Fernandez *et al.*, 2000; Lancellotti & Vasquez, 2000). Major upwelling events do not occur as they do in northern Chile, although small, sporadic, coastal upwelling occurs in the summer months (January to March) in the north of the Valdivia region (Moreno *et al.*, 1998). These events do not contribute substantially to regional differences in nearshore production of phytoplankton, with productivity related more clearly to land use (Van Holt, 2009) and nutrients primarily delivered by terrestrial inputs (Hebbeln *et al.*, 2000; Van Holt, 2009).

We examined 13 different watersheds subject to a variety of changes in upstream land use. Mechanized agricultural activities in the valleys include production of fodder (64%) and cereals such as barley, wheat and oats (17%) [Insituto Nacional de Estadísticas (INE), 1997]. Dairy farms in the valleys and mountain foothills account for 1.5 million cows, which are raised on grasses and fodder (INE, 1997). Native forests primarily remain in difficult to reach areas in the Andes and Coastal Mountains (CONAF *et al.*, 1997). The 'Decreto Ley 701' law of 1974 subsidized 75% of the cost to establish new plantations for pulp (Lara & Veblen, 1993). Plantations of pines and eucalypts have expanded, and Chile is a leading exporter of wood for pulp (Sedjo, 1999). We compared large watersheds (4800–15 300 km^2) and smaller watersheds (265–1390 km^2), with each watershed numbered and named according to its main river (Fig. 1). The watersheds in the *Northern Valdivian* (watersheds #1, 2, and 3) and *Valdivian* (watershed #4) zones contain extensive tree plantations and agriculture. The *Osorno* zone (watersheds #5, 6, 7, 8, 9, and 10; Fig. 1) is characterized by extensive native forest cover, agriculture, and an increasing, number of plantations. Watersheds in the *Llanquihue* zone (watersheds #11, 12, and 13) support agriculture and ranching, as well as *matorral* (a term used for both new growth and shrubby forest) and a few plantations. Urban and commercial activities are not common in any of the zones, with the exception of the Celulosa Arauco (CELCO) pulp mill located in the *Valdivian* zone that began operating in late 2003. Each zone contains all or part of a major city i.e., Valdivia, Osorno or Puerto Montt (Fig. 1). The national census carried out in 2002 recorded approximately 140 000 people in Valdivia and Osorno and 320 000 or a little more than double that amount in Puerto Montt (INE, 2002). The *Northern Valdivia* and *Valdivia* zones exhibited higher chlorophyll-*a* values than either the *Llanquihue* or *Osorno* zones, with Van Holt (2009) reporting that changes in chlorophyll-*a* values at the downstream ends of the watersheds were positively related to the relative (%) and absolute (km^2) amount of the watershed converted to plantation between 1985 and 2001.

Sampling and satellite image processing

Between June and August of 2003 (the *loco* harvest season), a total of 1374 *locos* were collected for analysis. Thirty *locos* were taken from each of the 42 management areas (Fig. 1). We sampled primarily legal-sized *locos*, i.e., *locos* that were $\geq$ 10 cm long. We sampled additional smaller *locos* (N = 114) when possible. All samples were frozen until they could be processed. In the laboratory, wet weights of muscle tissue (WGHMT), viscera (WGHV), and shells (WGHSH) were measured to 0.1 g with an Ohaus HH320 balance. Shell lengths (SHELG) and heights (SHELHGT) were measured with vernier calipers. The ratio of WGHMT to SHELG was calculated as a fat index (FATIND) to identify *locos* that were considered 'skinny' (i.e., those with relatively low relative weight given their shell size). Such animals are known locally as *locos flacos*, and they are considered undesirable by fishers.

An acetate sheet marked with 25 randomly selected sampling points was used to quantify percent cover of endobionts and epibionts associated with *locos* sampled from the various management areas. The transparency was placed on the inner and outer surfaces of designated shells, and we counted epibionts and endobionts in contact with each of the 25 points, including barnacles, polychaetes, chitons, limpets, bryozoans, juvenile *locos*, green and red algae, snails, nudibranchs, amphipods, isopods, hydrozoans, mussels, and tunicates. Counts were multiplied by four to scale them to 100. *Locos* throughout the study area bore a number of endobionts; most notably *Phoronis ovalis*, the native *Dodecaceria* cf. *opulens* polychaete (Moreno *et al.*, 2006), and the shell-boring bivalve *Hiatella solida* (hereafter referred to as phoronids, polychaetes, and bivalves). *Verruca laevigata* and *Jehlius cirratus barnacles,* hereafter referred to as barnacles, were observed to be the primary epibionts. In our final analysis, we focused on occurrences of phoronids (PPHOR), polychaetes (PPOLYC), bivalves (PBIVAV), and barnacles (PBARN). Phoronids (*P. ovalis*) and polychaetes (*Dodecaceria* cf. *opulens*) bore into *locos*, and barnacles colonize the top of the shell (Arenas, 1972; Gallardo & Osorio, 1977; Moreno *et al.*, 2006; Avilés *et al.*, 2007). In our samples, polychaetes, barnacles, and phoronids were the only organisms that averaged over 5% cover across all management areas. Bivalves were observed less frequently (~1% average cover), but historical records of their occurrence had been collected by Gallardo & Osorio (1977), and fishers expressed concern about the effect of these bivalves. Other epibionts and endobionts were too rare for inclusion in analyses.

Two of the largest, two of the smallest, and two medium sized *locos* from each management area were selected for analyses of lipid content (PLIP) as an additional indicator of condition (Sibly & Calow, 1986). Lipids were measured as described by Folch *et al.* (1957).

For the Land Cover/Land Use Change (LCLUC) analysis, we used an October 5, 1985 Landsat 5 TM and a November 29, 2001 Landsat 7 ETM+ scene (WRS II path 233, rows 87–89) in combination with a maximum likelihood supervised classification to create nine landscape classes: (1) native forest, (2) plantation, (3) *matorral*, (4) agriculture, (5) cleared land, (6) urban, (7) wetland, (8) water, and (9) snow. Images were georectified and had a root mean square error less than 15 m. We applied an atmospheric correction using the Center for the Study of Institutions, Populations, and Environmental Change (CIPEC) protocol that involved converting radiance values to

surface reflectance values and dark object subtraction (Markham & Barker, 1986; Teillet & Fedosejeus, 1995). We also converted the low-gain portion of band 6 to temperature values. All scenes were topographically corrected using the Minnaert Correction (Minnaert, 1941), with modification as described by Teillet *et al.* (1982). For the supervised classification we stacked multiple layers that included all Landsat bands plus: a tasseled-cap transformation that converted bands to six new axes that represent brightness, greenness, wetness, fourth, fifth, and sixth layer (Kauth & Thomas, 1976); a normalized differential vegetation index layer (NDVI) (Rouse *et al.*, 1974); three texture layers that included range, mean, and standard deviation of band 5; and a slope image derived from a Shuttle Radar Topography Mission (SRTM) Finished 3 arc-second (90 m) raster elevation dataset.

We used the Chilean vegetation map (CONAF *et al.*, 1997) and our knowledge of the spectral signatures to select training points. We collected 109 accuracy samples from July to August in 2004 with the assistance of a Chilean forest engineer who helped characterize the vegetation characteristics of each site.

Tree plantations were visually obvious on the images, but difficult to separate solely by their spectral signatures. We restricted the tree-plantation analysis because of spectral confusion with disturbed native forest and because of the large size of the study area (35 853 km^2). We created a vector layer that included a 1 km buffer around plantation polygons from the national vegetation map (CONAF *et al.*, 1997), which is based on aerial photos, and supplemented this with visually obvious plantation regions that we identified. We used the preliminary classification of 1985 and 2001 to further refine the classification. For example, we double-checked regions that had cleared land in 1985 and native forest in 2001.We reclassified those plantation pixels (usually those pixels were inside areas already classified as plantations) and left the correct native forest pixels. We also sieved the urban class and made a minimum urban class consist of 10 pixels using an 8-neighbor rule; this eliminated confusion between cleared land and urban areas. We ran two accuracy assessments; one based on accuracy points selected in the field and another based on randomly selected points and ancillary land cover maps.

Chlorophyll-*a* concentrations were extracted from SeaWiFS satellite images (9 km resolution) using Giovanni (Acker & Leptoukh, 2007). Only two raw SeaWiFS satellite images were cloud free during the study period; therefore, the Giovanni images were determined to yield the best approximation of chlorophyll-*a* values throughout the study region. For each 0.1 degree latitude, the chlorophyll-*a* values (mg m^{-3}) across one degree longitude were averaged for April, May, June, and July from 1999 to 2003 to create the chlorophyll-*a* variable (CHLA) (mg m^{-3}) for each management area. This time period was chosen to minimize contributions from upwelling. Upwelling is not a major source of nutrients within our region of interest, and available literature suggests that only minor upwelling events occur during these months (Moreno *et al.*, 1998). The 0.1 degree latitudinal extent corresponded well to each management area. Average chlorophyll-*a* concentrations were chosen to represent the onshore to offshore decreases observed consistently.

Analyses

Distribution maps for the following variables: PPOLYC, PBIVAV, PPHOR, PBARN, PLIP, and WGHMT were created and combined to evaluate regional patterns in the condition of *locos*. Condition variables for *locos* (WGHMT, SHELGT, SHELHGT, PLIP, PPHOR, PPOLYC, PBIVAV, and PBARN) were analyzed further with ANOVAS, cluster analyses, and canonical correlation analyses. FATIND, SHELWDT, and WGHV were excluded from the analyses as they were highly correlated with the other variables and their introduction into the models resulted in unacceptable multicollinearity. Chlorophyll-*a* in the management area (CHLA) (mg m^{-3}) was used for the discriminant function and canonical correlation analyses. The LCLUC variables percent agricultural change (PAG_CH), percent cleared land change (PCL_CH), percent urban change (PURB_CH), percent change in *mattoral* (PMAT_CH), absolute areal (km^2) change for plantations (PLCH), and percent plantation change (PPL_CH) were used in discriminant function and canonical correlation analyses.

Differences in *loco* condition characteristics (WGHMT, SHELGT, SHELHGT, PLIP, PPHOR, PPOLYC, PBIVAV, and PBARN) across the three geographic zones, i.e., Valdivia (*Northern Valdivia* and *Valdivia*), *Osorno*, and *Llanquihue*, were assessed with a one-way ANOVA followed by a multiple comparison test ($\alpha = 0.05$) using Least Square Means with the Bonferroni correction (*P*-value adjusted for number of comparisons) (SAS, SAS/STAT; SAS Institute Inc., Cary, NC, 2001).

A non-hierarchical cluster analysis followed by a hierarchical cluster analysis was used to group sites according to *loco* condition rather than geography. For *loco* condition, an average value for the WGHMT, SHELGT, SHELHGT, PLIP, PPHOR, PPOLYC, PBIVAV, and PBARN, was computed for each of the 42 management areas. The non-hierarchical analysis provided the most robust clusters and the hierarchical analysis provided the classification tree (McGarigal *et al.*, 2000). The non-hierarchical cluster analyses used a polythetic, non-overlapping, sequential agglomerative method to cluster the variables based on the correlation matrix, which facilitated elimination of more highly collinear variables from subsequent steps (SAS, SAS/STAT; SAS Institute Inc.). This was followed by a hierarchical cluster analysis that generated relationship trees, with Ward's minimum variance linkage yielding clusters that best matched the groupings of the initial non-hierarchical clustering.

Next, a stepwise discriminant analysis was run using the groupings of management areas obtained from the hierarchical cluster analysis as categories and the condition characteristics as explanatory variables to identify which condition variables (WGHMT, SHELGT, SHELHGT, PLIP, PPHOR, PPOLYC, PBIVAV, and PBARN) best discriminated the clusters (SAS, SAS/STAT; SAS Institute Inc.). Another step-wise discriminate analysis was run using the groupings of management areas as categories and landscape change (PAG_CH, PCL_CH, PURB_CH, PMAT_CH, PL_CH, and PPL_CH) and phytoplankton biomass characteristics (CHLA) as explanatory variables.

A canonical correlation analysis identified combinations of condition variables that were highly correlated with a combination of LCLUC and chlorophyll-*a* measures for management areas (SAS, SAS/STAT; SAS Institute Inc.). The squared canonical correlation statistic indicated how strongly canonical variables were related, and the standardized canonical coefficients identified which *loco*, LCLUC and chlorophyll-*a* variables contributed the most to each canonical pair. The LCLUC variables tested included: (PAG_CH, PCL_CH, PURB_CH, PMAT_CH, PL_CH, and PPL_CH) and chlorophyll-*a* in the management area (CHLA). As measures of condition, we included WGHMT, SHELGT, SHELHGT, PLIP, PPHOR, PPOLYC, PBIVAV, and PBARN.

We used small *locos* that were sampled opportunistically to understand patterns in the growth of the shellfish not captured in the cross-sectional study design. *Locos* from northern Chile (27°S) have been shown to grow slower than *locos* found further south (32°–47°S), with a maximum difference in shell length of 20 cm reported in *locos* of the same age (Moreno, 1997).

Results

Landscape change

The 1985 classification had overall accuracies of 91% and 90%, with kappa coefficients of 0.90 and 0.89, for the field-based and randomly-generated accuracy assessments, respectively. The 2001 classification had overall accuracies of 88% and 92%, with kappa coefficients of 0.86 and 0.91, respectively. Since 1985, 1992 km^2 of plantations were established within the study region (Van Holt, 2009). The greatest percentage of change occurred in the Lingue watershed (#1); plantation cover increased by 151 km^2 resulting in a total plantation cover changing from 30% to 35%. The Valdivia watershed (#3) exhibited the largest areal increase (1264 km^2), which translated into a 12% increase in tree plantations since 1985 and a 2001 total percentage cover of 14%. Across the study site, the land converted to tree plantations came in nearly equal amounts from native forest and agricultural lands (Van Holt, 2009). Native forest cover remained high in many smaller watersheds along the coastal mountains (83–92%).

Chlorophyll-a patterns

When grouped by geographic zones, the *Northern Valdivia* zone had highest mean chlorophyll-*a* values in its management areas (2.5 ± 0.3), followed by the *Valdivia* zone (2.5 ± 0.1), the *Llanquihue* zone (1.4 ± 0.2), and the *Osorno* zone (1.1 ± 0.1). The mean chlorophyll-*a* concentration across all management areas was 1.8 ± 0.6 mg m^{-3}.

Geographic patterns in the condition of locos

Higher percentages of phoronids, barnacles, and polychaetes were found on *loco* shells located near watersheds with extensive or numerous tree plantations, i.e., those in the *Northern Valdivia* and *Valdivia* zones (Figs 2 and 3). The mean percentage cover of phoronids on *loco* shells in the *Northern Valdivia* region was highest (27.8%) and statistically different ($P < 0.001$) from records for the other zones (0.0–1.7%) (Figs 2 and 3). *Locos* in the *Northern Valdivian* zone had significantly higher percentage cover of polychaetes (8.1%) than shellfish in the *Llanquihue* zone (5.0%; $P = 0.045$). The percentage cover of polychaetes on *Valdivian locos*, however, was statistically similar ($P > 0.634$) to coverage on shellfish from all other zones. The *locos* from the *Northern Valdivia* and *Valdivia* zones had statistically higher ($P < 0.001$) percentage covers of barnacle epibionts (45.8% and 48.2%, respectively) compared to *locos* from the *Osorno* and *Llanquihue* zones (15.3% and 21.2%). In contrast, the percentage cover of bivalves was statistically higher ($P \leq 0.007$) in the *Osorno* zone (2.8%). *Locos* in *Northern Valdivia*, *Valdivia*, and *Llaquihue* zones had relatively low (0.7%, 1.5%, and 0.6%) and statistically equal ($P > 0.341$) coverages of bivalves. The Least Significant Difference (LSD) analysis indicated that the mean tissue weight of *locos* from the *Osorno* zone (90.1 g) was less than that for shellfish

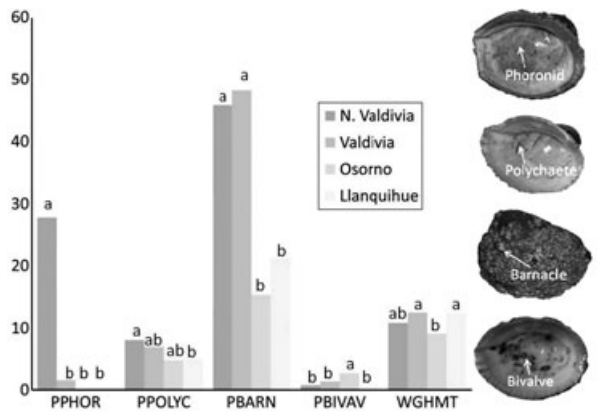

Fig. 2 Mean values for *loco* shells used in ANOVA and multiple comparisons based on geography (management areas corresponding to Fig. 1). The letters correspond to similar groups (based on a multiple comparison test with a Bonferroni correction where $\alpha = 0.05$).Also *loco* shells showing (a) phoronid, (b) polychaete, (c) barnacles and (d) bivales. (a), (b), and (d) are showing the ventral side of the *loco* shell. (c) is the dorsal side. Phoronids leave light brown marks on shell (a). The dark black lines in (b) are polychaetes. The circular marks on shell (d) reflect the damage from the bivalves. The y axis represents percentage cover for all variables except WGHTMT, which is in decagrams.

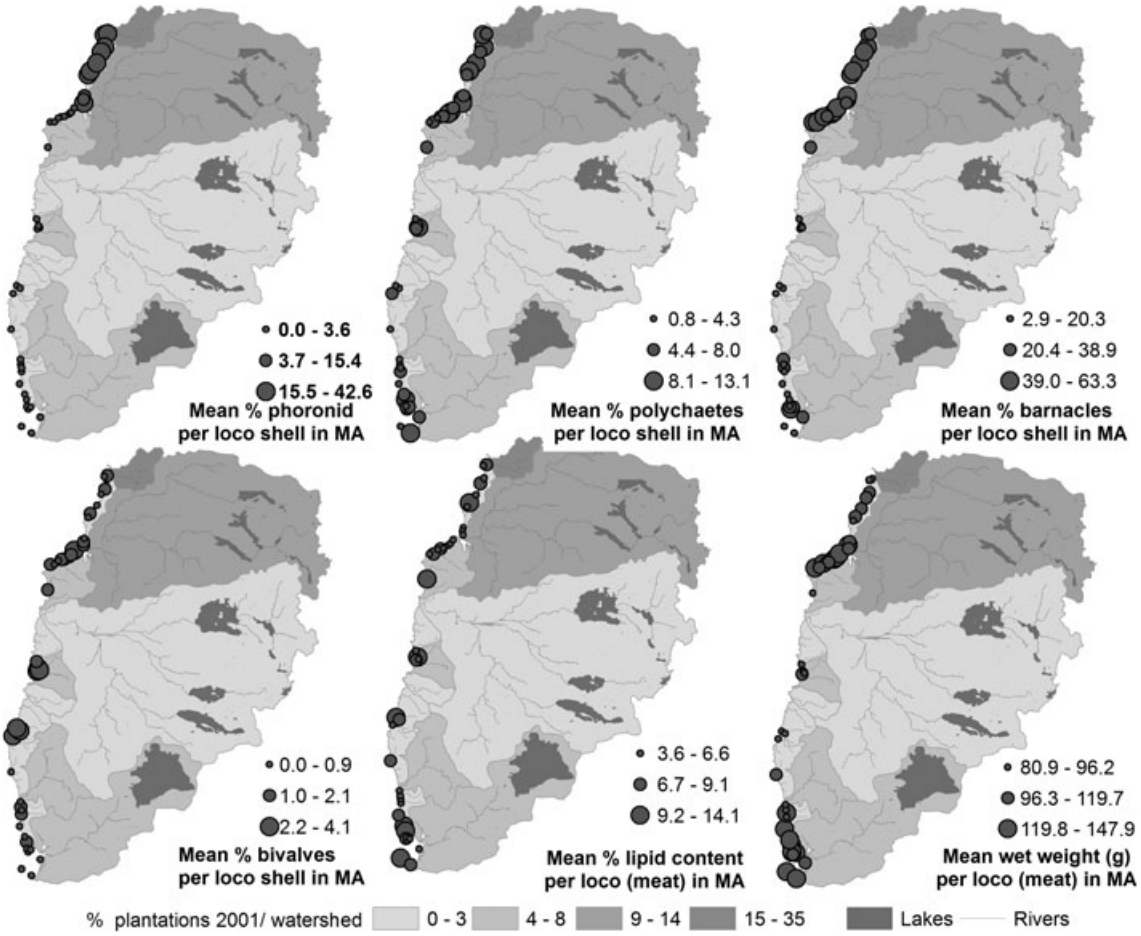

Fig. 3 Average percent (a) phoronid, (b) polychaete, (c) barnacle, and (d) bivalve cover of on *loco* shells as well as (e) lipid content and (f) weight of *loco* meat in each management area and the percent forest plantation cover in 2001. (Averages obtained from a sample of 30 *locos* per management area).

from the *Valdivia* (124.9 g; $P < 0.001$) and *Llanquihue* zones (123.3 g; $P < 0.001$). *Locos* from the *Northern Valdivia* zone were statistically equal ($P = 0.158$) in size to those from the *Osorno* zone, although *Osorno locos* were slightly heavier (108.4 g). *Locos* from the *Northern Valdivia* zone also yielded tissue weights that were statistically equal to those recorded for *locos* from the *Llanquihue* and *Valdivia* zones ($P > 0.146$), but the former *locos* weighed slightly less.

Cluster analysis of loco condition

Four distinct groups of *locos* (Fig. 4) were identified based on measures of their condition. *Locos* from management areas that were geographically close together often were in the same group. The *Northern Valdivia* group contained *locos* from management areas from north of Rio Valdivia. The *Valdivia* group included *locos* from management areas in the Valdivia province (below and a few above Rio Valdivia), but it also included *locos* from a few management areas (Estaquilla B, Amortajado, and Cullinco) in the *Osorno* and *Llanquihue* zones. The *Osorno* group contained *locos* from the *Osorno* zone, with the exception of Estaquilla and Guar Guar, which actually are located in the *Llanquihue* zone. Within the *Llanquihue* group, three subgroups were identified: two of which contain management areas located in the *Llanquihue* zone, and another that had *locos* from all zones.

Groups of *locos* were discriminated primarily by percentage cover of phoronids. Percentage cover of bivalves was the next most important variable in discriminating groups, and it was followed by heights of shells, weights of *loco* tissue, percentage cover of barnacles, and percentage cover of polychaetes (Table 1). Shell length and percent lipid content for *loco* meat were not significant discriminators. *Locos* from *Northern Valdivia* had at least five times more phoronids than any other group, *Osorno locos* had the most bivalves (3% total cover), and *locos* from *Northern Valdivia*

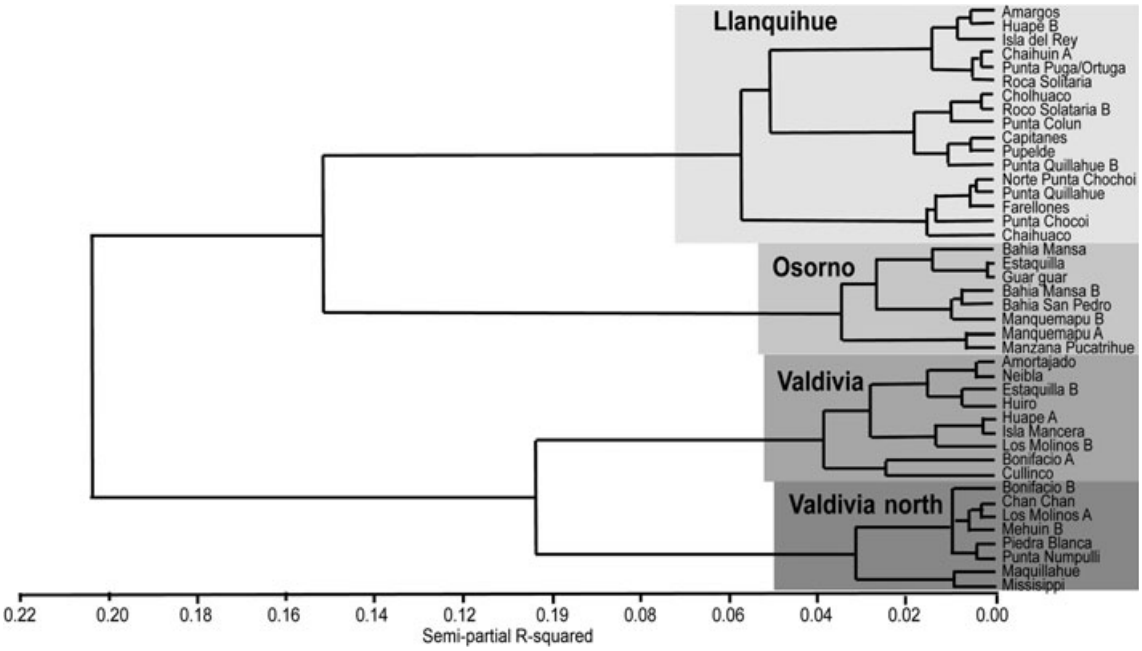

Fig. 4 Cluster analysis of *loco* shellfish in each management area based on condition (percent phoronid, polychaete, and bivalve cover on *loco* shells, as well as weight and lipid content of meat, and shell length). Distinct levels of gray signify distinct clusters.

and *Valdivia* had at least 1.5 times more barnacles than those collected from *Llanquihue* and *Osorno*. *Loco* meat from *Llanquihue* and *Valdivia* weighed the most, about 10–30 g more than meat from *locos* taken in *Valdivia North* and *Osorno* (see Table S1 for mean values of *loco* cluster group characteristics).

Loco condition clusters and discriminant analysis of LCLUC variables

The discriminant analysis based on *loco* groups identified percent change in plantations from 1985 to 2001 (PLCH) as the variable that best explained the variation among groups, with chlorophyll-*a* concentrations (CHLA) being the next best predictor (Table 1). The misclassification rates provide opportunities to understand more fully why *locos* from some management areas apparently fared better or worse than expected given the land cover and chlorophyll-*a* concentrations. The overall misclassification rate from the discriminant analysis was 37%, but misclassifications were unevenly distributed among groups (e.g., see Tables S2 and S3 for misclassification estimates and probabilities). *Osorno* and *Northern Valdivia*, which contained *locos* with extreme condition measures and high values for LCLUC, had the lowest misclassification rate (12%). In contrast, *Valdivia* had

Table 1 Partial R-square, significance levels, and average squared canonical correlations from the stepwise discriminant analysis of A.) *loco* grouping using mean values for phoronids (PPHOR), shell-boring bivalves (PBIVAV), polychaetes (PPOLYC), barnacles (PBARN), shell height (SHELLHT), *loco* meat weight (WGHMT), shell length (SHELLGT) and percent lipid in *loco* meat (PLIP). and B.) LCLUC characteristics using percent change in plantations from 1985 to 2001 (PPL_CH) and ocean color characteristics (CHLA) in a canonical discriminant analysis of the hierarchical cluster analysis

Analysis	Step	Variable	Partial R-Square	Pr > F	Average squared canonical correlation
A.	1	PPHOR	0.8199	<0.0001	0.273
	2	PBIVAV	0.6728	<0.0001	0.488
	3	SHELLHT	0.5033	<0.0001	0.620
	4	WGHMT	0.3200	0.0034	0.672
	5	PBARN	0.2225	0.0339	0.703
	6	PPOLYC	0.2085	0.0497	0.737
B.	1	PLCH	0.5917	<0.0001	0.197
	2	CHLA	0.3370	0.0015	0.293

89% of the management areas misclassified, and *Llanquihue* had a 35% misclassification rate. Cross-validation indicated that by using just CHLA and PCPL one could expect an overall misclassification rate of 43% (see Table S4 for cross-validation). Thus, other factors associated with certain management areas appeared to influence the condition of *locos*.

The LOCO1 canonical variable measures '*loco* condition' primarily because the variables PPHOR, PBARN, WGHMT, and PBIVAV all had high weights (Table 2). The LCLUC1 canonical variable was interpreted as a measure of 'plantation development' primarily because two variables, the percentage of plantation change and chlorophyll-*a* levels, have high positive weights (see Table S5 for canonical variables). The squared canonical correlation for LOCO1 and LCLUC1 was 0.881 and a scatter plot of these factors demonstrated a linear and highly significant ($P < 0.0001$) relationship (Fig. 5). A high canonical correlation between LOCO1 and LCLUC1 suggests that *locos* with more phoronids and barnacles are associated with management areas that have high chlorophyll-*a* values in the coastal waters, which, in turn, relates to nutrient inputs to nearshore waters with adjacent tree plantations.

All other canonical measures were difficult to interpret because the factor weights did not follow consistent patterns. *Loco* lipid content and *loco* meat weight were not important factors in distinguishing *loco* groups.

Loco length and epibont/endobiont relationships

Landscape-level patterns likely trump the influence of age on percentage covers of phoronids and polychaetes.

Table 2 Standardized canonical coefficients for the three canonical variables, LOCO1, LOCO2 and LOCO3 that describe the association between *Loco* Health Variables and *loco* groupings. The LOCO1 canonical variable measures '*loco* health' primarily because the variables PPHOR, PBARN, WGHMT, and PBIVAV all have high weights

Loco variables	LOCO1 Epibionts & endobionts	LOCO2	LOCO3
WGHMT	0.253	0.452	−0.173
SHELLGT	−0.188	0.424	0.212
SHELLHT	−0.069	−0.478	0.066
PBARN	0.438	−0.130	0.950
PPOLYC	0.193	0.234	0.073
PPHOR	0.675	−0.030	−0.770
PBIVAV	0.240	−0.680	0.236
PLIP	−0.032	0.041	0.076

PPHOR, phoronids; PBIVAV, shell-boring bivalves: PPOLYC, polychaetes; PBARN, barnacles; SHELLHT, shell height; WGHMT, *loco* meat weight; SHELLGT, shell length; PLIP, percent lipid in *loco* meat.

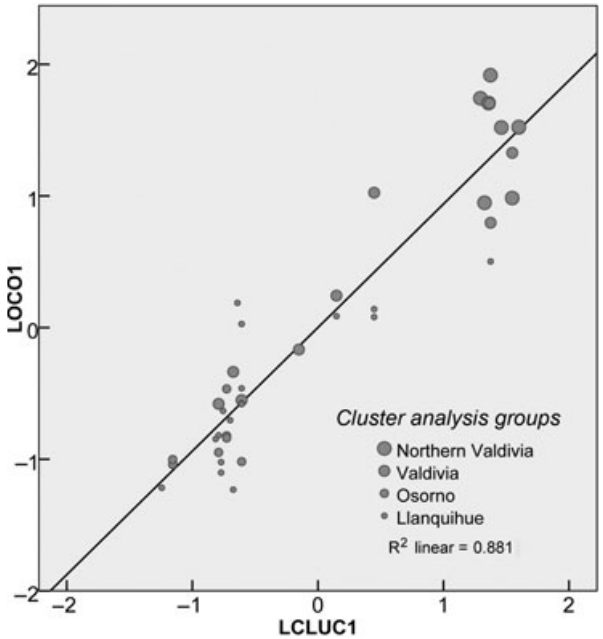

Fig. 5 Scatterplot of LOCO1 (predominantly endobiont phoronids and epibiont barnacles) on *loco* shells and LCLUC1 (predominantly plantation development and chlorophyll-*a* variables) from the canonical correlation analysis ($R^2 = 0.881$; $P = 0.000$). The size of the circle corresponds to the groups created with the hierarchical cluster analysis (Fig. 4).

Locos from regions influenced by plantations (derived from ANOVA and cluster analyses) had an average of ~3.5% more polychaetes and 23% more phoronids than *locos* from other regions. The maximum variation in size was 50 mm across all groups. A regression model of *loco* length (independent) vs. epibionts and endobionts (dependent) for all *locos* ($N = 1324$) show that a 50 mm increase in shell length resulted only in 1.5% more polychaetes and 3.0% fewer phoronids ($Y = 0.03x–2.08$, $R^2 = 0.02$; $Y = –0.06x + 8.9$, $R^2 = 0.01$, respectively).

Bivalves accounted for less than 5% of the coverage on *loco* shells, and variation in infestation with age may explain a large fraction of the variation in bivalve cover. Most bivalves were found on *locos* from regions least influenced by landscape change, and *locos* in the regions with more bivalves were on average larger than other *locos*. Regions with more bivalves had ~3% more bivalve coverage than areas with few bivalves (Fig. 2). A regression indicated that a 50 mm increase in *loco* size resulted in 1% more bivalves (1/3 of the 3% bivalve coverage), so some other factor also is affecting coverage ($y = 0.02x–1.9$; $R^2 = 0.05$).

Overall, higher percentage covers of polychaetes and bivalves were associated with longer (older) *locos*, when compared to smaller (younger) *locos*. Coverage of phoronids showed the opposite effect, *loco* length

diminished as phoronids increased. Barnacles were present on both small *locos* and large *locos*.

Discussion

Landscape change thresholds and fishery effects

More land was converted to tree plantations in the *Northern Valdivia* and *Valdivia* zones than in other regions; chlorophyll-*a* concentrations in the adjacent coastal waters were correspondingly higher; and barnacle, polychaete, and phoronid infestations were apparent. Located in the *Northern Valdivian* zone, the Lingue and Valdivia watersheds had the largest percentage (14–35%) and areal (180–1400 km^2) cover of tree plantations. The influence of the tree plantations translated to approximately a 26% increase in phoronids, 3% more polychaetes, and a 24–33% increase in barnacle cover on *locos*. This increase would result in a $2–$3 reduction in price per kilogram of *loco* meat (Van Holt, 2012). This translates into a 29–43% reduction in the price reportedly paid to fishers. In contrast, the Bueno watershed had 490 km^2 of tree plantations in 2001, but since the Bueno watershed is so large, this accounts for only 3% of the total cover and *loco* shells were cleaner. The Bueno watershed is within the *Osorno* zone which contains more extensive stands of native forest that likely served as a buffer mitigating the influence of the tree plantations on the nearshore system.

Although most of the plantation development occurs in watersheds with significant amounts of agriculture, trees are not being planted on prime agricultural lands (Van Holt, 2009) but rather on sloped hillsides considered marginal for agriculture. Steeper slopes increase the potential for run-off and erosion that will influence nearshore systems. Agriculture is likely not playing the defining role in the nutrient input into the system because higher chlorophyll-*a* values and poorer quality locos are near the Lingue watershed, which is losing agricultural land, and the Valdivia watershed, which had less than 1% increase in agriculture. The Bueno watershed, on the other hand, had relatively healthy *locos* despite a 5% increase in agriculture, which in this large watershed, translated into an 836 km^2 increase from 1985 to 2001 and a total agricultural area of 5403 km^2 in 2001. Another possibility worth exploring is that the tree plantations in the *Osorno* zone are derived from mainly afforested agricultural areas, which may play a role in minimizing the effect on nearshore systems. In the *Northern Valdivia* and *Valdivia* zones plantations are more frequently replacing native forests. More research is needed to isolate the effects of slope, land cover, and history of land use in order to tease apart the interactions between distinct landscape-level patterns of development and the consequences of establishing tree plantations.

Barnacle epibionts and phoronid and polychaete endobionts were good indicators of sites affected by landscape change. Bivalve presence, on the other hand, was not positively related to landscape change, and it was not considered a good indicator. Meat weight and lipid content measures may be useful indicators, but their associations with landscape change and chlorophyll-*a* variables were weaker. Phoronids, if prevalent, may negatively affect *loco* length. Our data indicate that barnacles are early colonizers of *loco* shells, and they are followed by polychaetes and phoronids. Gallardo & Osorio (1977) hypothesized that the presence of some epibionts and endobionts (barnacles and phoronids) fosters the growth of other epibionts and boring organisms (bivalves). Although we found no evidence of this phenomenon with shell-boring bivalves, we observed a significant positive correlation between barnacle cover and phoronid and polychaete cover (see Table S6 for correlations among health condition variables). Bers & Wahl (2004) showed that substrate microtopography influences epibiont presence and Wahl (1989) explained that the presence of some epibionts is dependent on the presence of another. We argue that barnacles might be a necessary successional stage that allows abundant colonization of phoronids and polychaetes. Barnacles may offer some protection from wave action and predators, or perhaps create flows that direct food toward phoronids or polychaetes in an advantageous way. Sites with heavy barnacle cover and few phoronids and polychaetes might represent an opportunity to distinguish the effects of nutrients from other influences, although this is hypothetical.

Historical records for the *Northern Valdivia* zone in Chile show that epibiont and endobiont presence has increased on *locos* since the 1970s as have tree plantations. In the 1970s, *locos* had high levels of phoronid infestation (100 phoronid holes per cm^2) which is similar to the coverage seen today (Arenas, 1972). Gallardo & Osorio's (1977) study, however, demonstrated that not all *locos* were infested. Only three *locos* (14% of sampled *locos*) had phoronids present and the infestation was not as extensive as reported by Arenas (1972) because the shells did not exhibit layer thickening that is present with massive infestations of *P. ovalis*. Gallardo & Osorio (1977) also reported that polychaetes and barnacles were also less abundant (only 43% of the *loco* shells analyzed had these organisms). Our samples at the same location in 2003 show *all* (100%) of the *locos* had phoronids, barnacles, and polychaetes. Our *locos* were heavily infested with phoronids and the

thickening of shell resulted in a narrower shell width. In addition, phoronid infested *locos* in our study were shorter than uninfected animals.

Our findings agree with research that associates barnacles, polychaetes, phoronids, and bivalves with eutrophication across the world. Belan (2003), for example, found more barnacles in eutrophic sites in the Amursky Bay in the Sea of Japan. Gabaev *et al.* (2005) found high abundance of barnacle larvae in eutrophic sites in the Amur Bay in the Sea of Japan. They argued that barnacle abundance was related to diatom abundance, which is associated with eutrophication, since diatom prey accounted for ~90% of all phytoplankton. Chang *et al.* (1992) and Belan (2003) found more species and higher abundances of polychaetes and phoronids associated with increased nutrients in the New York Bight and the Amursky Bay in the Sea of Japan, respectively. Llanso (1992) showed that opportunistic polychaetes dominated in times of hypoxia in the Lower Rappahannock River in the Chesapeake Bay, and Beukema (1991) noted polychaete dominance in the tidal-flat benthic community of the Dutch Wadden Sea after 20 years of increased nutrients. Perus & Bonsdorff (2004) reported that polychaetes accounted for the top five most dominant species in a 7 years study of eutrophication in archipelagos and coastal areas in the Baltic Sea. Crema *et al.* (1991) found large abundances of polychaetes associated with eutrophication in northern Adriatic Sea. Risk *et al.* (1995) were not able to provide conclusive evidence about the relationship between nutrients and bivalves, however, fewer bioeroding bivalves were observed at offshore reef sites in the Great Barrier Reef (less runoff), whereas sites closest to nutrient runoff were less affected than those at a moderate distance from the nutrient source. These global accounts, however, are not linked to changes in landscapes or transitions in land use at the local level.

The availability of prey for *locos* likely exerts an additional level of control on how intensely *locos* are affected by phoronids, polychaetes, and barnacles (Gallardo & Osorio, 1977). Fewer food resources could influence the *loco*'s ability to combat epibionts by limiting the *loco's* ability to produce antifouling agents. Although it is purely hypothetical that *locos* produce antifouling agents, they are produced by other species and it is energetically costly to do so (Wahl, 1989). Research prior to the establishment of management areas in the *Northern Valdivian* zone has shown that mytilids, which *locos* consume, were abundant in the region and at times dominant because tourists collected other more valuable species, whereas in the marine reserve, where *locos* were protected from harvest, the mytilids were not quite as abundant (Jara & Moreno, 1984; Moreno *et al.*, 1984, 1986). Although the establishment of the management areas may have reversed the dominance of mytilids, Moreno's (2001) study showed that potential prey assemblages in the *Northern Valdivia* system were less diverse than in other zones.

The misclassified observations in the discriminant analysis offer evidence supporting the idea that sufficient food resources may mitigate the effects of nutrient inputs. The Bonifacio A management area was classified in the *Valdivia* group in the hierarchical cluster analysis using *loco* characteristics. However, using the LCLUC characteristics, this site would be classified as belonging to *Northern Valdivia*. Bonifacio A had *locos* with better condition indices than one would expect for its LCLUC score and is considered by local fishers to be one of the best sites in *Northern Valdivia*, always producing abundant, good quality *locos*. Bonifacio A *locos* were probably classified in *Valdivia* because *locos* there had fewer phoronids than *locos* in Bonifacio B. The Bonifacio B management area, which is adjacent to Bonifacio A and managed by the same group of fishers, was not as productive. It is possible that unique hydrographic conditions support more prey for *locos* in Bonifacio A, which diminishes the *locos'* susceptibility to phoronids. J. Zamorano, the scientific advisor to the fishers in the Bonifacio region, and C. Moreno (personal observation) have both observed anomalous eddy-like features near Bonifacio A, and they note that mytilids (Bivalvia: Mytilidae) and tunicates (*Pyura chilensis*) are common at this location. Bonifacio B, in comparison, harbors fewer mytilids and tunicates. Indeed *locos* from Bonifacio A had unexpectedly high lipid content, which also could be the result of more available prey.

The suspension feeding barnacles, phoronids, and polychaetes would have reduced the amount of phytoplankton in the system (Herman & Scholten, 1990); however, this trophic link was likely weaker than a release from top-down control as the health of *locos* declined. Reyes & Moreno (1990) measured annual growth on a sample of *locos* that were marked and recaptured for 9 years. *Locos* from the Mehuin region in Chile (near Lingue watershed) that were affected by epibionts and endobionts showed truncated growth after 5 years (see Figure S1 for models of *loco* length by age), with negative effects on growth and biological condition. If narrower width and truncated growth reduce predation by *locos*, then the abundances of filter-feeding organisms should increase, which should reduce phytoplankton standing crops and potentially lead to a negative feedback on the occurrence of epibionts and endobionts. In conditions without excessive anthropogenic inputs of nutrients

and subsequent increases in phytoplankton, such feedback mechanisms should help keep *loco* populations healthy and stable.

Consequences for the fishery

Phoronid-affected *locos* were slightly smaller in length and narrower in width. Because the price of *locos* is based primarily on *loco* length and weight, *locos* that weigh less than predicted based on a given length *(locos flacos*) result in lower prices paid to fishers. Van Holt (2012) has shown that the price fishers reportedly are paid for their catch is influenced by polychaete and phoronid cover. Because the management system currently in place restricts fisher movement, fishers are limited in how they can adapt to changes in shellfish quality. When the condition of the *locos* is poor, fishers harvest in unassigned areas outside of their management area, go further out to sea and harvest other species, or leave the fishery altogether (Van Holt, 2012). For example, in the Mehuin region of Chile, an area that has poor quality *locos*, fishers are leaving the fishery because of the low-quality resources and the conflict with the Celco Arauco pulp mill (Van Holt, 2012). Plantations are likely to increase in southern Chile because many areas near current plantations are cleared or in the early stages of establishment (Van Holt, 2009). Thus, *locos* in other regions may be subject to changes in condition as a consequence of landscape change. An integrated coastal management approach is necessary to mitigate the influence of epibionts and endobionts, particularly phoronids, barnacles, and polychaetes on *locos* and the fishery. More broadly, integrative coastal models need to link large-scale landscape transformations, such as tree plantations, nearshore fisheries and the people that depend on both sources of income.

Conclusions

Recovery of the Chilean loco fishery is attributed to an innovative management plan developed and implemented in the 1990s. Sustainability of the fishery, however, is potentially compromised by the indirect effects of land-use on the quality of the locos' nearshore coastal habitat. The development of tree plantations within watersheds at or near the land-sea interface results in increased chlorophyll-*a* concentrations in adjacent nearshore waters (eutrophication) and promotes the growth of fouling organisms (suspension feeding endobionts and epibionts) that, in turn, affect the biological condition of *locos*. The relatively poor condition of *locos* in waters adjacent to watersheds with extensive tree plantation cover are a primary concern for fishers and fisheries resource managers as the value of harvested organisms is significantly reduced. Findings reported herein highlight an important linkage between human land-use activities and ecological processes in the nearshore coastal environment, and they indicate clearly that more integrated approaches to natural resource management are needed, with socio-economic trade-offs associated with specific land uses explicitly identified and incorporated into decision making processes.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Figure S1. Loco length (mm) by age. Mehuin data obtained from Reyes & Moreno (1990). Ancud data obtained from Bustos *et al.* (1986). After 5 years loco growth in Mehuin gets truncated. Ancud data is not based on growth rings, rather than annual mark and recapture, the curves can be compared to see the standard growth and truncated growth. Locos in Ancud have few epibionts/endobionts.

Table S1. Mean and standard deviation (in parenthesis) values of *loco* characteristics of each group in the hierarchical cluster analysis for phoronids (PPHOR), shell-boring bivalves (PBIVAV), polychaetes (PPOLYC), barnacles (PBARN), shell height (SHELLHT), *loco* meat weight (WGHMT), shell length (SHELLGT) and percent lipid in *loco* meat (PLIP)

Table S2. Number of observations, percent correctly classified (diagonal) and misclassified (off diagonal) estimates of hierarchical cluster analysis groups *Llanquihue*, *Osorno*, *Valdivia*, and *Northern Valdivia* using percent change in plantations from 1985 to 2001 (PLCH), percent native forest in 2001 (PFNF) and chlorophyll-*a* (CHLA) characteristics from canonical discriminant analysis.

Table S3. Probability of group membership of misclassified observations of the canonical discriminant analysis of hierarchical cluster analysis groups *Llanquihue*, *Osorno*, *Valdivia*, and *Northern Valdivia* using percent change in plantations from 1985 to 2001 (PLCH), percent native forest in 2001 (PFNF), and chlorophyll-*a* characteristics (CHLA).

Table S4. Cross-validation summary and error count estimates of the canonical discriminant function of hierarchical cluster analysis groups *Llanquihue*, *Osorno*, *Valdivia*, and *Northern Valdivia* using percent change in plantations from 1985 to 2001 (PCPL) and chlorophyll-*a* characteristics (CHLA).

Table S5. Standardized canonical coefficients for the three canonical variables, LCLUC1, LCLUC2, and LCLUC3 that describe the importance of each landscape change variable on the canonical variable. The LCLUC1 canonical variable is interpreted as a measure of 'plantation development' primarily because the percentage of plantation change and chlorophyll-*a* levels have high positive weights. Note land cover change is from 1985 to 2001 and PAG_CH = percent change in agriculture, PCL_CH = percent cleared land change, PURB_CH = percent urban land class change, PMAT_CH = percent mattoral change, PL_CH = area (km^2) plantation change, and PPL_CH = percent plantation change. CHLA = chlorophyll-*a* characteristics in the management area.

Table S6. Pearson's Correlation among *loco* health condition variables where WGHMT = weight of *loco* meat (g), WGHV = weight of *loco* viscera (g), WGSH = shell weight (g), SHELGT = shell length (cm), SHELHGT = shell height (cm), SHELWDT = shell width (cm), FATIND = fat index (WGHMT/SHELGT), PLIP = percent lipid in *loco* meat, PPOLYC = percent polychaete cover on shell, PPHOR = percent phoronid cover on shell, PBIVAV = percent bivalve cover on shell; PBARN = percent barnacle cover on shell; MALE = sex, male coded as 1. Note: ** is highly significant where $P < 0.01$ and * is significant where $P < 0.05$.

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